

Supplementary Material

Cohort characteristics

Supplementary Table S1 provides a cohort-level description of the analytic sample used in this study. The in-domain set comprises 2,803 subjects from 12 cohorts, while the out-of-domain evaluation set comprises 1,304 subjects from 4 held-out cohorts. For each cohort, we report the sample size, diagnostic composition, sex distribution, age at MRI acquisition, and number of acquisition sites.

Table S1. Cohort-level composition of the analytic sample. For each cohort, we report sample size, class distribution across the seven diagnostic categories, sex distribution, age at MRI acquisition (mean $\pm$ SD and range), and number of acquisition sites. Totals are reported separately for the ID and OOD sets. Across cohorts, T1-weighted MRI acquisitions included both 1.5T and 3T scanners.

Dataset	N	Diagnosis (<i>n</i>)							Sex F/M	Age (years) mean $\pm$ SD [range]	Sites
		AD	CN	DLB	bvFTD	PNFA	SD	PSP			
In-domain (ID)											
ADNI	736	332	404	–	–	–	–	–	354/382	75.0 $\pm$ 6.8 [55.0–91.0]	60
CamCAN	442	–	442	–	–	–	–	–	222/220	65.2 $\pm$ 12.3 [44.2–88.9]	1
IXI	319	–	319	–	–	–	–	–	193/126	60.9 $\pm$ 9.2 [44.0–86.2]	3
ALLFTD	308	5	–	1	154	41	39	68	134/174	65.9 $\pm$ 7.4 [49.0–83.0]	28
SALD	282	–	282	–	–	–	–	–	179/103	58.7 $\pm$ 8.8 [44.0–80.0]	1
DLBS	209	–	209	–	–	–	–	–	131/78	66.5 $\pm$ 12.7 [44.3–89.1]	1
OASIS	192	45	147	–	–	–	–	–	135/57	69.2 $\pm$ 13.6 [44.0–96.0]	1
PDBP	168	–	22	125	–	–	–	21	47/121	68.8 $\pm$ 7.2 [50.0–90.0]	13
ICBM	70	–	70	–	–	–	–	–	40/30	56.3 $\pm$ 8.6 [44.0–80.0]	1
MIRIAD	69	46	23	–	–	–	–	–	38/31	69.4 $\pm$ 7.0 [55.6–85.8]	1
ABIDE	5	–	5	–	–	–	–	–	1/4	47.7 $\pm$ 5.0 [44.1–56.2]	n/a
NDAR	3	–	3	–	–	–	–	–	0/3	46.9 $\pm$ 2.9 [44.2–49.9]	n/a
ID total	2803	428	1926	126	154	41	39	89	1474/1329	67.1 $\pm$ 11.1 [44.0–96.0]	–
Out-of-domain (OOD)											
NACC	682	441	161	47	20	6	7	–	413/269	73.1 $\pm$ 9.9 [30.1–100.2]	42
AIBL	279	47	232	–	–	–	–	–	149/130	72.3 $\pm$ 7.1 [55.0–93.0]	2
NIFD	276	–	135	–	70	34	37	–	132/144	63.6 $\pm$ 7.3 [39.0–81.0]	3
4RTNI	67	–	–	–	–	–	–	67	38/29	70.7 $\pm$ 7.4 [55.0–86.0]	4
OOD total	1304	488	528	47	90	40	44	67	732/572	70.8 $\pm$ 9.5 [30.1–100.2]	–